

CoRoT-2b

R-0146

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_180601 · created 2026-07-16T08:35:17+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458270.8345 +/- 0.0022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.122 +/- 0.023
Transit depth (Rp/Rs)^2	1.48 +/- 0.55 [%]
Orbital Inclination (inc)	86.83 +/- 2.0
Airmass coefficient 1 (a1)	0.998 +/- 0.02
Airmass coefficient 2 (a2)	0.0004 +/- 0.0043
Scatter in the residuals of the lightcurve fit is	1.98 %
Best Comparison Star	#2 - [84, 167]
Optimal Aperture	2.78
Optimal Annulus	11.09
Transit Duration (day)	0.0858 +/- 0.0079

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.122 ± 0.023 vs 0.1667 (deviation 1.91σ)
Duration fit vs published	0.0858 d vs 0.095 d (deviation 1.14σ)
Mid-time O–C	4.3 min
Depth SNR	5.2
Residual scatter	1.98 %

Light curve

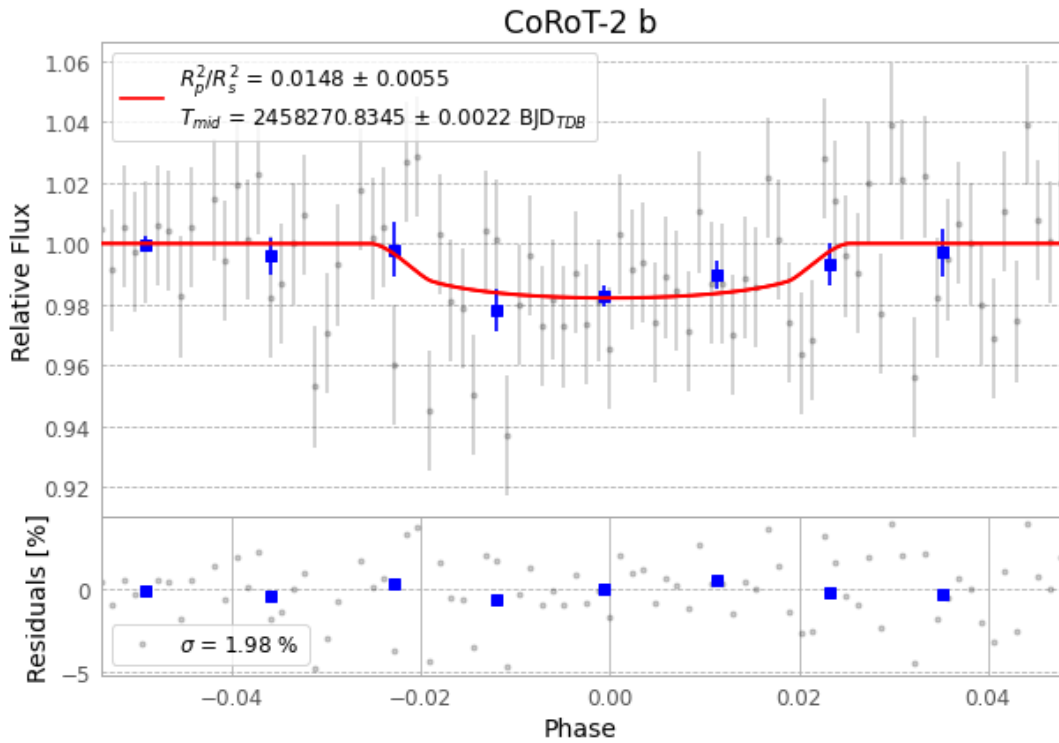


Figure 1 — FinalLightCurve_CoRoT-2 b_2018-05-31.png (SHA-256 f638034cbc94ea23...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [343, 164], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 68c8fd3dd9153403...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 3ebc8ba

Data provenance

86 FITS frames (56 MB), CoRoT-2180601053912.FITS → CoRoT-2180601095412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-180601.log (46dcad5a3049...), FinalLightCurve_CoRoT-2 b_2018-05-31.png (f638034cbc94...), FinalLightCurve_CoRoT-2 b_2018-05-31.csv (63121cb09171...)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 08:35Z.